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Review article

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Dimensionality engineering of perovskites for stable heterojunction-based photovoltaics

Randi Azmi[1], Drajad Satrio Utomo[2], Yanping Liu[1], Shynggys Zhumagali[2] & Stefaan De Wolf[2]

Abstract

Commercial solar cells require long-term operational stability. Despite their high performance, perovskite solar cells degrade owing to defects, impurities and mobile ions in the bulk and at the surface of their photo-absorbing 3D metal-halide perovskite films. Compared with 3D perovskites, low-dimensional (LD) perovskites exhibit greater phase stability and superior ambient, light and thermal stability. Notably, by forming 3D/LD heterostructures, these LD layers can also passivate defective 3D perovskite surfaces through surface reconstruction. However, this approach can increase energy mismatch and structural disorder at the contact interfaces owing to excess unbonded ligands. The LD perovskite capping layers can also feature mixed phases, random orientations and other inhomogeneities, which can create charge recombination channels, jeopardize charge transport and undermine long-term stability. Moreover, the monovalent ammonium-based ligands (phenethylammonium and butylammonium) commonly used to create 3D/LD heterojunctions are relatively unstable owing to weak van der Waals interactions btween the organic sheets and the inorganic framework, as well as their relatively low acid dissociation constant (pK_a), which make them prone to deprotonation. To improve stability, it is thus imperative to use suitable organic ligands that form strong coordination bonds with the inorganic framework – ideally multivalent amines with high pK_a values. Here, we review instability mechanisms at 3D/LD interfaces and discuss mitigation strategies, focusing on ligand chemistry and the fabrication of phase-pure, homogeneous LD capping layers to improve 3D/LD perovskite heterostructure stability.

Sections

[1]School of Science and Engineering, Chinese University of Hong Kong (Shenzhen), Shenzhen, P. R. China. [2]Center for Renewable Energy and Storage Technologies (CREST), King Abdullah University of Science and Technology (KAUST), Thuwal, Saudi Arabia. e-mail: randiazmi@cuhk.edu.cn; stefaan.dewolf@kaust.edu.sa

Introduction

The performance of 3D metal-halide perovskite solar cells (PSCs) has seen considerable improvement since their first demonstration in 2009. However, their intrinsic instabilities under long-term operation remain a key barrier for their commercialization, owing to their high defect density, ion migration at grain boundaries and inhomogeneous distribution of defects or impurities (including residual PbI_2 and the non-perovskite yellow phase) at the surfaces of perovskite films[1–6]. To mitigate this instability, a common strategy involves post-treating the top surface of the 3D perovskite film by applying a solution containing organic ammonium-based ligands, which reacts with surface defects to form low-dimensional (LD) perovskites or 0D, 1D or 2D perovskites[1,7–14]. The resulting heterostructure, consisting of a 3D/LD bilayer structure featuring either a continuous or discontinuous LD capping layer, serves as one of the most effective strategies for passivating the surface of 3D perovskites (Fig. 1a). Besides providing chemical and field-effect passivation, this approach enhances the structural properties and can improve energy-level alignment with the charge-selective contacts, thereby simultaneously reducing interfacial recombination and improving charge extraction[1,7–10,15–26]. As a result, the dimensional heterojunction method stands out as one of the most promising surface engineering approaches to enhance device performance (Fig. 1b and Supplementary Table 1).

Despite this success, this approach often suffers from stability variations owing to the presence of excess unbound ligands and the formation of a poorly controlled mixture of phases with random orientations, especially for 2D perovskites. These factors lead to interfacial energy mismatch and structural disorder, which detrimentally affects the chemical and physical properties of the 3D/LD heterostructure[1,4,10,18,20,27–29]. Besides, monovalent ammonium ligand spacers with low steric hindrance (such as phenethylammonium (PEA) or *n*-butylammonium, which are commonly used to form 2D Ruddlesden–Popper perovskites) only exhibit weak chemical interactions between the organic ligand sheets, which can disrupt the octahedral network within the 2D perovskite matrix and promote diffusion of the organic ligands into the 3D bulk film[29–38]. This process is accelerated under actual device operating conditions, usually entailing elevated temperatures and intense illumination[29–32,38,39]. Given their detrimental impact on device performance, these factors mandate systematic and intensive investigations. Insufficient understanding often results in poorly controlled formation of 3D/LD heterostructures, which in turn leads to failures in long-term stability and may cast doubt on the effectiveness of the 3D/LD heterostructure approach in general[6,27,32–34,36,38,40].

In this Review, we systematically survey the mechanisms underlying the degradation of 3D/LD heterostructures based on commonly used monovalent ammonium-based spacers and solution fabrication processes. We also discuss available strategies to counteract the degradation of 3D/LD heterostructure and elucidate their specific advantages and limitations. These strategies include designing ligand spacer chemistry for both monovalent and multivalent ammonium-based ligands to enable the formation of a more robust octahedral network for LD perovskite or perovskitoid layers, as well as implementing fabrication engineering approaches to construct a high-quality LD capping layer with highly ordered structures, phase purity and improved quality. With this Review, we aim to strengthen confidence in the 3D/LD heterostructure strategy and support its ability to meet industrially accelerated stability tests, thereby advancing the commercialization of this approach in photovoltaics.

Tuning the structure and properties of low-dimensional perovskites

LD perovskites and perovskitoids are generally obtained by reducing the connectivity of the octahedral inorganic network $[PbX_6]^{4-}$ using an organic ligand inserted through corner-sharing, face-sharing or edge-sharing, or a combination of these modes[41–44] (Fig. 2a). For instance, in LD perovskitoid layers, organic ligands can connect to the octahedral network through edge-sharing and face-sharing connections, or in combination through corner-sharing. These diverse bonding modes offer greater structural diversity and high stability[12,45–48]. By contrast, 2D perovskites are obtained by connecting organic ligands to the inorganic octahedral network through corner-sharing only, but they benefit from more organic ligand options owing to preferable coordination[41,49]. In 1D perovskites, the $[PbX_6]^{4-}$ octahedra form highly distorted, coplanar chains with linear connectivity. These chains are surrounded by organic ligands that interact through face-sharing or edge-sharing, resulting in structures with the general formula $LPbX_3$ or L_3PbX_5 (refs. 11,50,51), respectively. One-dimensional perovskites typically extend over 1 µm in length. A further reduction in the connectivity of the inorganic network leads to 0D perovskites or perovskite nanocrystals. This structural isolation allows greater flexibility in how organic ligands coordinate, as there is no direct connection between the inorganic units. Zero-dimensional or nanocrystal perovskites typically have sizes greater than 100 nm.

These ligand arrangement variations offer flexibility in designing LD layers with enhanced material functionalities and electronic properties and open new pathways for designing materials with tailored dimensionalities.

Two-dimensional perovskites can generally be expressed as $L_2A_{n-1}B_nX_{3n+1}$ (for the Ruddlesden–Popper phase) and $L'A_{n-1}B_nX_{3n+1}$ (for the Dion–Jacobson phase), in which L and L′ denote monovalent aromatic or aliphatic and divalent ammonium-based ligands, respectively. The Ruddlesden–Popper phase is formed by interdigitating two monovalent ligands, connected through van der Waals interaction. Each ammonium group from a monovalent ligand substitutes an A-cation site and connects with the octahedral inorganic network through hydrogen and/or coordination bonding[5,9,17,52–55]. By contrast, for the Dion–Jacobson phase, two octahedral inorganic sheets are connected through one divalent ligand with ammonium groups on both ends, eliminating the van der Waals gap. Besides these two types, the alternating cation interlayer phase, which mostly uses guanidium cations[51], has mixed Ruddlesden–Popper and Dion–Jacobson features. Each phase offers distinct advantages in terms of stability and optoelectronic properties (Supplementary Table 2). Briefly, Ruddlesden–Popper phases provide strong moisture resistance owing to their hydrophobic organic spacers, but the weak van der Waals bonding and wide insulating layers limit charge transport and structural robustness. Dion–Jacobson phases, by contrast, eliminate the van der Waals gap through divalent linkers, resulting in smaller interlayer spacing, stronger structural stability and improved charge transport, although they are generally less hydrophobic. Alternating cation interlayer phases combine features of Ruddlesden–Popper and Dion–Jacobson structures, offering intermediate interlayer distances, stability and charge transport, but their long-term behaviour remains less understood[3,28,56–58].

On the basis of the aforementioned discussion, dimensional heterojunction engineering can be used to construct LD capping layers with different electronic, structural, physical and chemical properties[1,6,8,9,16–19,21,29,59,60]. These tailored LD capping layers enable 3D/LD heterostructure perovskites to be stabilized through different

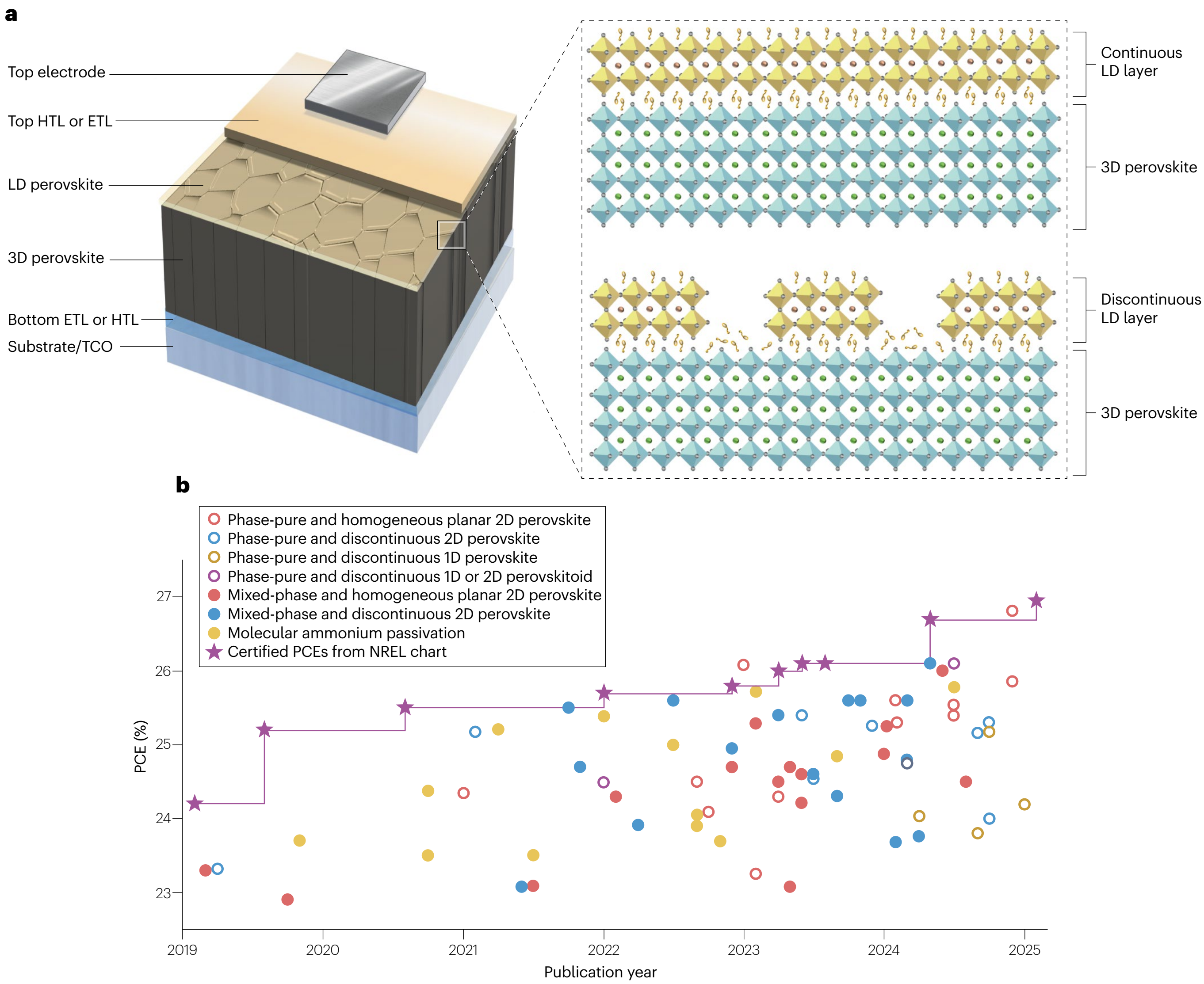


Fig. 1 | Dimensional heterojunction perovskite device architecture and performance progress. a, 3D/low-dimensional (LD) heterojunction device architecture at the top contact with continuous and discontinuous 2D perovskite capping. **b**, Summary of power conversion efficiencies (PCEs) of perovskite solar cells with various types of LD passivation layers considering their distribution and dimensionality purity (Supplementary information). ETL, electron transport layer; HTL, hole transport layer; TCO, transparent conductive oxide.

mechanisms. For instance, 2D perovskites can be synthesized with varying dimensionalities and different n value ($n = 1, 2, 3, \ldots$), which enables the modulation of the quantum well and energetic levels, that is, the conduction band minimum (CBM), valence band maximum (VBM) or work function[61–63] (Fig. 2b). Accurate energy-band alignment at the heterojunction interface is of prime importance to improve device performance and can be further modulated by combining two or more layers. There are two types of heterojunctions in 3D/LD heterostructure perovskites according to their energetic alignment configurations. For type I heterojunctions, the CBM of the 3D perovskite is lower than the CBM of the LD perovskite, whereas the VBM of the 3D perovskite is higher than the VBM of the LD perovskite. In this case, the electrons or holes from the 3D perovskite cannot be transferred or moved efficiently to the LD layer[8,17,18,29,61,64]. By contrast, for type II heterojunctions, the CBM of the 3D perovskite is higher than the CBM of the LD perovskite, and the VBM of the 3D perovskite is lower than the VBM of the LD perovskite, creating a staggered band structure between the 3D and the LD perovskite[9,17,20,21,29,64,65]. Most 2D perovskites have a relatively higher VBM than that of the 3D perovskite, resulting in more efficient charge transport at the perovskite/hole contact[9,18,21,66]. However, very few 2D perovskites have a lower CBM than that of 3D perovskites[8,17,20,29,48,65]. For this reason, further interface-dimensionality engineering needs to be explored to achieve type II band alignment, which would enable effective 2D passivation at the electron-selective contact side while enhancing electron transfer at the perovskite/electron-selective contact interface[8,20,29].

Several methods have been introduced to address charge transport and mobility at the heterojunction interface[1,8,9,16,17,20,25,64,67]. Specifically,

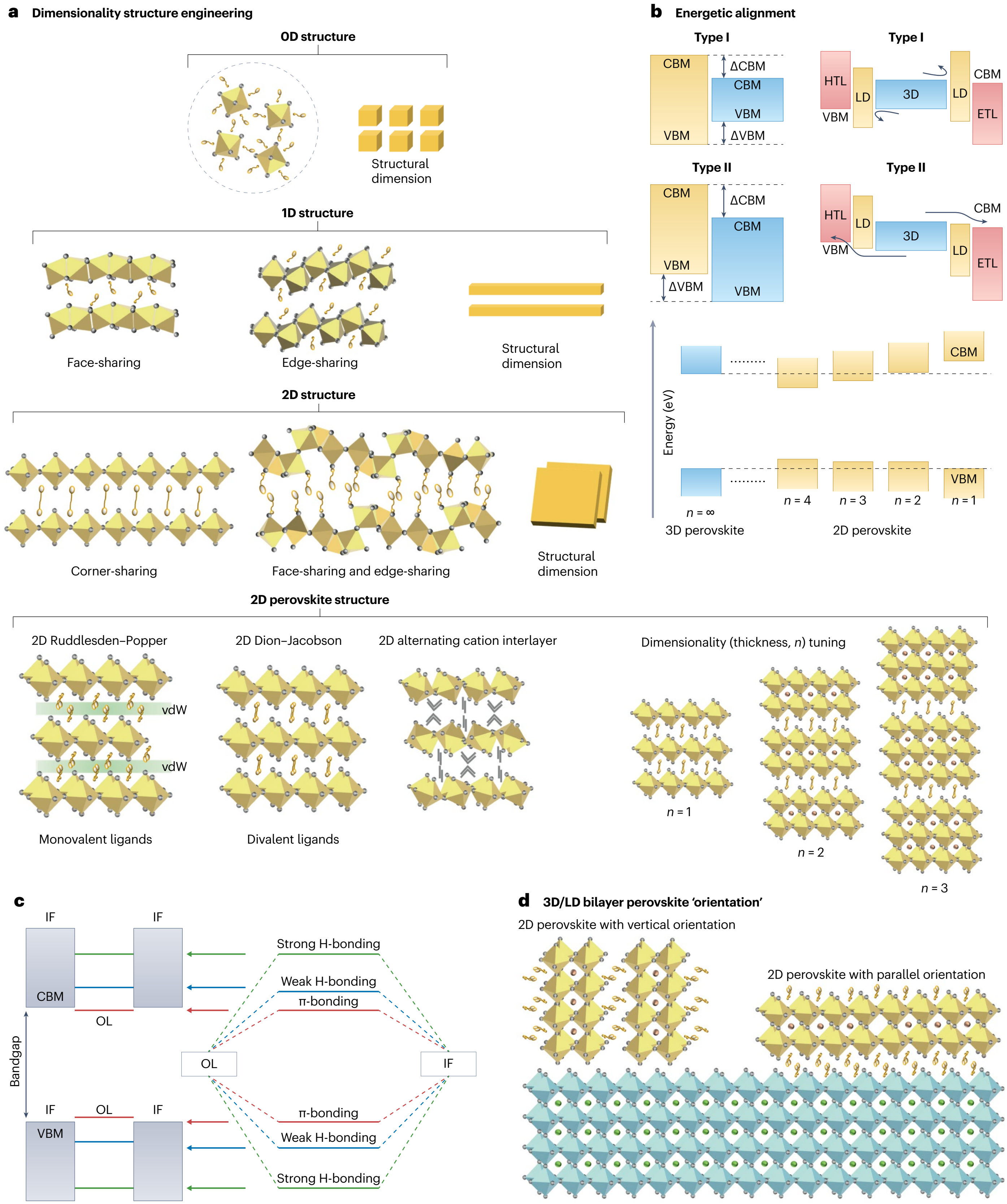

a Dimensionality structure engineering
0D structure
Structural dimension
1D structure
Face-sharing
Edge-sharing
Structural dimension
2D structure
Corner-sharing
Face-sharing and edge-sharing
Structural dimension
2D perovskite structure
2D Ruddlesden–Popper
vdW
vdW
Monovalent ligands
2D Dion–Jacobson
Divalent ligands
2D alternating cation interlayer
Dimensionality (thickness, n) tuning
n = 1
n = 2
n = 3
b Energetic alignment
Type I
CBM
VBM
ΔCBM
CBM
VBM
ΔVBM
Type I
HTL
VBM
LD
3D
LD
CBM
ETL
Type II
CBM
VBM
ΔVBM
ΔCBM
CBM
VBM
Type II
HTL
VBM
LD
3D
LD
CBM
ETL
Energy (eV)
CBM
VBM
n = ∞
n = 4
n = 3
n = 2
n = 1
3D perovskite
2D perovskite
c
IF
IF
CBM
OL
Bandgap
IF
OL
IF
VBM
Strong H-bonding
Weak H-bonding
π-bonding
OL
IF
π-bonding
Weak H-bonding
Strong H-bonding
d 3D/LD bilayer perovskite 'orientation'
2D perovskite with vertical orientation
2D perovskite with parallel orientation

Fig. 2 | Compositional engineering of LD perovskite layers and properties. **a**, Structures of low-dimensional (LD) derivates, including 0D nanocrystal or 'dot' perovskites, 1D 'rod' perovskites and 2D 'sheet' perovskites (either in the Ruddlesden–Popper, Dion–Jacobson or alternating cation interlayer phase). Dimensionality tuning in 2D Dion–Jacobson perovskites ($n = 1$, 2 and 3). **b**, Types of heterojunction band alignment and band alignment between 3D perovskites and 2D perovskites of different bandgaps and dimensionalities. The dashed lines guide to the valence band maximum (VBM) and conduction band minimum (CBM). **c**, Band offsets between the inorganic frame (IF) and the organic ligand (OL) sheet under different bonding processes. **d**, High-phase-purity 3D/2D heterojunction with varied orientations of the 2D perovskite layer. ETL, electron transport layer; HTL, hole transport layer; vdW, van der Waals.

for heterojunctions containing 2D perovskites, the transport of charge carriers in the out-of-plane direction is impeded by the high resistance associated with the organic ligand sheets. To mitigate this issue, shorter functionalized monovalent or divalent ammonium-based ligands can be used[6,8,9,22,23,29,67,68]. Modulating chemical bonding between the organic ligands and the inorganic octahedral network can adjust the band offset between these adjacent layers (Fig. 2c). Large monovalent organic cations can be substituted by shorter divalent cations, which effectively reduce the spatial distance between inorganic slabs and improve the out-of-plane charge mobility[9,20,23]. Additionally, stronger hydrogen bonding between organic ligands and inorganic layers leads to a deeper VBM of the 2D perovskite layer, resulting in a higher barrier for hole extraction[9]. Reducing the strength of the hydrogen bonding in the organic ligands can result in a much lower energetic barrier for hole extraction leading to better energetic alignment[9]. Conversely, weaker hydrogen bonding can jeopardize the integrity of the 2D perovskite layer; hence, a fine balance must be established for efficient charge transport at the 2D/3D heterojunction[9,17]. As another strategy, conjugated organic ligands that contain multiple aromatic rings forming extended π-bond networks can be used. These π orbitals can overlap to create continuous energy bands within the 2D perovskite layer, enabling more efficient electron transport[16]. This method is advantageous for 2D and other LD phase passivation at the electron-selective contact[16,20,29].

In the 2D perovskite layer, different preferential crystal growth orientations, either parallel or vertical to the 3D film, can be achieved to manipulate charge transport and mobility at the 2D/3D junction[17,18,20,21,29,67,69–71] (Fig. 2d). Vertical growth and a higher dimensionality of the 2D perovskite layer on top of the 3D perovskite are preferable for better charge transport[20,69,71]. However, this approach can often result in less stable 2D/3D perovskite structures compared with the parallel orientation, owing to ion mobility at the interface and the reduced hydrophobicity of the perovskite surfaces. This lower hydrophobicity can make the surface more susceptible to moisture-induced degradation[1,2].

The advantageous electronic and structural properties resulting from this combined approach underscore the potential of LD perovskite layers for designing perovskite thin-film heterojunctions. These properties contribute to improved control over electronic transport and internal electric fields and enable the achievement of tailored band alignment with specific band offsets. Realizing these advantages in actual devices, however, depends not only on the intrinsic design of LD phases but also on how they are incorporated into the perovskite structure.

Formation of 3D/LD heterostructures

To construct a 3D/LD perovskite heterostructure at the top-contact interface, solution post-treatment of the 3D perovskite film is commonly performed using spin-coating of a solution containing monovalent or divalent organic ligands, thereby forming in situ perovskite or perovskitoid capping layers with different structural phases onto the 3D perovskite film[1,4,6,9,17,19,20,27]. For the in situ formation of the LD layers, the primary underlying mechanisms involve the selective chemical reaction of the organic ligand on the surface of the 3D perovskite film, which occurs through cation exchange or through reaction with residual PbI_2 or impurity phases (such as δ-$FAPbI_3$) on the surface of the 3D perovskite film[1,4,20,23,27,72] (Fig. 3a).

Several chemical reactions can be used to form LD (either 0D, 1D and 2D) layers (Fig. 3b). The formation of 0D perovskites, either from cation exchange or reaction with residual PbI_2, requires extra halides, which can be added to the organic ligand solution. However, this requirement can complicate the fabrication process. Therefore, the in situ formation of the 0D phase has rarely been reported so far[12,73]. To solve this problem, sequential deposition of pre-synthesized 0D or nanocrystal perovskites can be performed through a solution process using nonpolar solvents such as toluene or octane, which do not damage the bulk 3D perovskite[74]. To enhance the formation of other LD phases such as 2D or 1D perovskites, an excess amount of PbI_2 (~5–15%) can be added to the 3D perovskite precursor solution, resulting in residual PbI_2 formation at the grain boundaries and surface of the 3D film[1,9,17,75]. Therefore, achieving controlled formation of specific LD phases requires precise regulation of the interaction between the organic ligands and these PbI_2 species.

Controlling the organic ammonium-based ligand reactivity and kinetic formation of the LD layer enables the obtention of a tailored distribution of LD capping layers, such as homogeneous planar or discontinuous structures[1,4,6,18,20,21,23,29,76–79]. Homogeneous planar structures refer to a continuous LD layer distribution on the surface of the 3D perovskite film, whereas discontinuous structures refer to an uneven LD layer distribution (Fig. 3b). This variation in LD layer distribution can be influenced by different factors, such as ligand reactivity, fabrication process, composition of the organic cations and the presence or distribution of excess PbI_2 on the surface of the 3D perovskite layer[1,18,21,23,29,37,67,71,78,80]. For instance, ligands with a lower reactivity towards the perovskite layer or a higher reactivity with PbI_2 are more likely to form discontinuous LD layers, owing to the minimal possibility for cation exchange reactions or to the uneven distribution of PbI_2 across the 3D perovskite surface[1,20]. For this reason, pre-depositing a uniform PbI_2 layer or choosing a ligand with high reactivity towards the perovskite layer can facilitate the formation of planar homogeneous LD layers[1,4,7,8,20,23,72]. A comprehensive understanding of the intricate reconstruction or reformation processes that occur during LD layer formation, driven by variations in organic ammonium-based spacers and fabrication methods, remains elusive yet critical for achieving controlled LD layer distribution[1,4,6,24].

For instance, smaller organic ligands with high reactivity can penetrate deeper into the 3D perovskite matrix, resulting in a higher dimensionality (higher n value, in which n represents the dimensionality of the LD perovskite by counting the number of its octahedral inorganic sheets), discontinuous or more infiltrated 2D layer distribution into the 3D film[8,23,26,79,81–83]. Zero-dimensional and 1D layers usually display discontinuous LD distributions, which might be due to geometrical

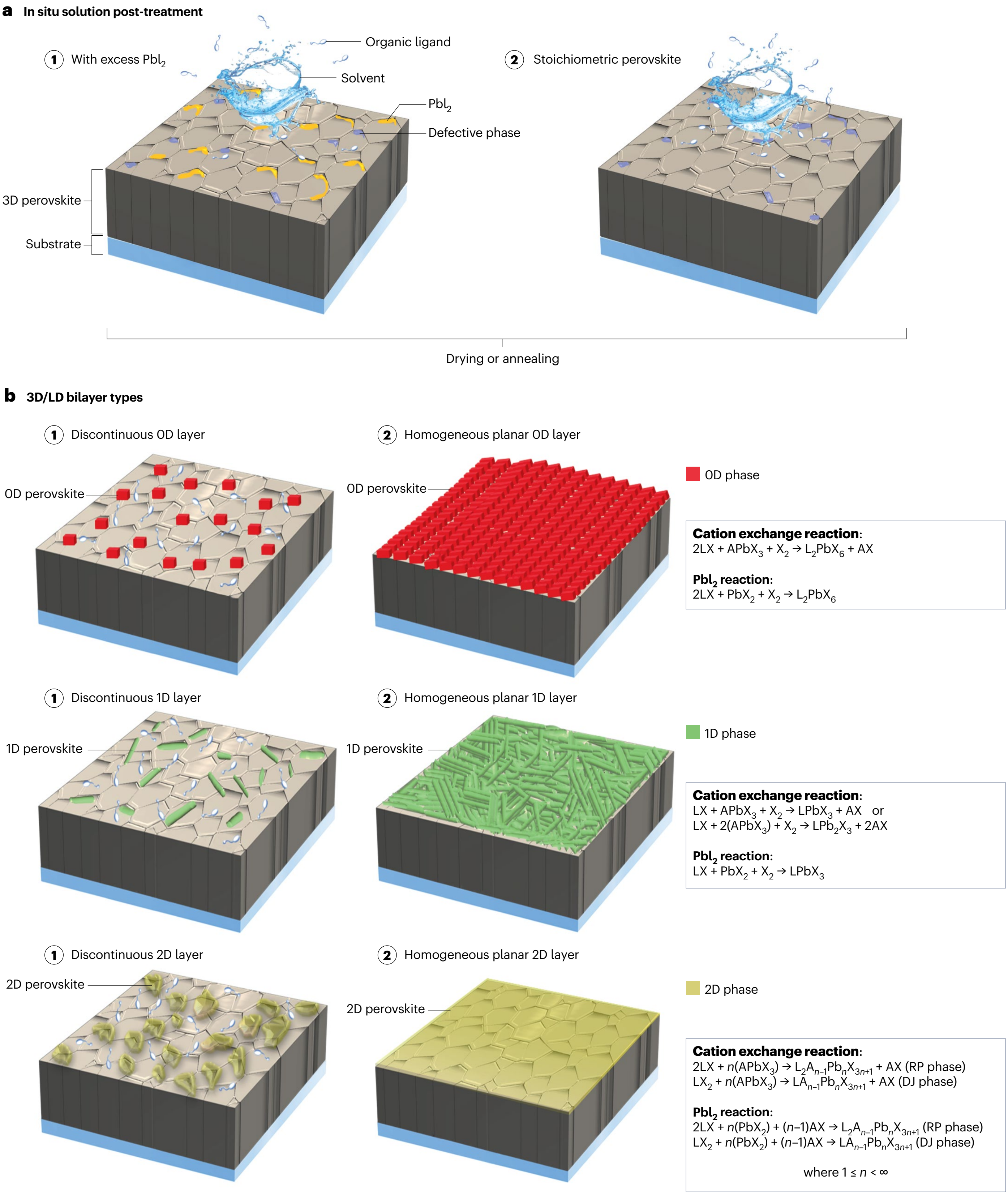

a In situ solution post-treatment
1 With excess PbI_2
Organic ligand
Solvent
PbI_2
Defective phase
2 Stoichiometric perovskite
3D perovskite
Substrate
Drying or annealing
b 3D/LD bilayer types
1 Discontinuous 0D layer
2 Homogeneous planar 0D layer
0D perovskite
0D perovskite
0D phase
Cation exchange reaction:
$2LX + APbX_3 + X_2 \rightarrow L_2PbX_6 + AX$
PbI_2 reaction:
$2LX + PbX_2 + X_2 \rightarrow L_2PbX_6$
1 Discontinuous 1D layer
2 Homogeneous planar 1D layer
1D perovskite
1D perovskite
1D phase
Cation exchange reaction:
$LX + APbX_3 + X_2 \rightarrow LPbX_3 + AX$ or
$LX + 2(APbX_3) + X_2 \rightarrow LPb_2X_3 + 2AX$
PbI_2 reaction:
$LX + PbX_2 + X_2 \rightarrow LPbX_3$
1 Discontinuous 2D layer
2 Homogeneous planar 2D layer
2D perovskite
2D perovskite
2D phase
Cation exchange reaction:
$2LX + n(APbX_3) \rightarrow L_2A_{n-1}Pb_nX_{3n+1} + AX$ (RP phase)
$LX_2 + n(APbX_3) \rightarrow LA_{n-1}Pb_nX_{3n+1} + AX$ (DJ phase)
PbI_2 reaction:
$2LX + n(PbX_2) + (n-1)AX \rightarrow L_2A_{n-1}Pb_nX_{3n+1}$ (RP phase)
$LX_2 + n(PbX_2) + (n-1)AX \rightarrow LA_{n-1}Pb_nX_{3n+1}$ (DJ phase)
where $1 \leq n < \infty$

Fig. 3 | Fabrication schemes for 3D/LD heterostructures. **a**, 3D perovskite film with PbI_2 and impurity defects fabricated using excess PbI_2 or stoichiometric precursor compositions, with post-treatment applied using spin-coating. **b**, Dimensional heterojunction types and corresponding synthetic pathways: discontinuous and continuous planar 0D (top), 1D (middle) and 2D (bottom) perovskite capping layers. DJ, Dion–Jacobson; LD, low-dimensional; RP, Ruddlesden–Popper.

limitations inherent in the arrangement of 0D or 1D phase structure over large areas[11–14,48,58]. By contrast, larger organic ligands with low or moderate reactivity and strong connections to the octahedral inorganic network typically form a more homogeneous planar 2D layer atop of the 3D film because of minimum ligand penetration[1,7,15,27,64,84]. However, homogeneous planar LD capping layers have been rarely reported owing to a limited understanding of their formation kinetics and of the role of the 3D perovskite surface composition[1,27,70]. Controlling surface reactions remains an important challenge owing to their inherent complexity, which complexifies the controlled growth of LD layers with optimal growth sites, morphologies and robust structural properties[4,20,27,48,72].

Solvent polarity, processing time and temperature may also impact the in situ formation of different LD layers on the 3D perovskite film[1,6,8,27,56,77,85]. Simply altering the concentration of the organic ligands in solution can regulate the thickness of the LD layer[1,8]. The choice of solvent to dissolve organic ligands also has a role in regulating the LD layer formation and distribution[1,8,27,77,85]. For example, polar protic solvents, such as isopropanol (IPA), and nonpolar solvents, such as chloroform, are commonly used to dissolve organic ligands[1,6,24,85]. Although chloroform can only dissolve alkyl-ammonium-based ligands, IPA can dissolve most organic ligands containing alkyl and aromatic groups. It is of note that IPA induces surface dissolution of the 3D perovskite layer owing to the solubility of its organic cations (formamidinium (FA) or methylammonium (MA)), leaving PbI_2 on the perovskite surface, which promotes the formation of an LD perovskite layer without requiring excess PbI_2 in the perovskite ink[8,22–24,82]. In addition, using a solvent mixture of IPA and dimethylformamide to deposit organic ligand results in the formation of a quasi-LD perovskite owing to partial dissolution of the perovskite upper layer, resulting in a mixed 3D/LD heterostructure with a non-distinctive junction[8,83]. Similarly, adding a co-solvent such as dimethyl sulfoxide in IPA facilitates the diffusion of organic ligands deeper into the 3D perovskite film[76,86]. Because of their high dielectric constant and Gutmann donor number, these co-solvents enable partial dissolution of the 3D perovskite upper layer, resulting in high dimensionalities with less homogeneity owing to the non-uniform evaporation of the solvents. For this reason, alternative co-solvents such as acetonitrile or ethyl acetate, which exhibit more uniform evaporation behaviour, may be used to improve 2D or 1D crystallization and enhance the permeability of the resulting film[76,87]. By contrast, chloroform has limited solubility for perovskites or their precursor salts. Treating the 3D perovskite with organic ligands in a chloroform solution requires the addition of excess PbI_2 into the perovskite precursor ink to alter LD film formation[1,26,85]. A lack of PbI_2 on the 3D surface can inhibit the formation of LD perovskites, leaving an excess of unreacted ligands. Thus, the choice of solvents influences the surface composition of the perovskite, resulting in variations in the quality of the LD capping layers[1,6,8,26,27,83,85].

The process of creating 3D/LD perovskite heterostructures typically also involves thermal annealing at a relatively moderate temperature (≤100 °C) for a few minutes[4,8,9,17,27,72,82,83]. A higher thermal annealing temperature (100 °C) might affect the 3D perovskite surface composition through surface reconstruction[29,65]. Although this mechanism can facilitate the formation of 2D perovskites at the 3D perovskite surface through enhanced kinetics, it typically results in structures with a low n value[1,22,24,26]. A prolonged thermal annealing process usually leads to the accumulation of excess PbI_2 species that can hinder charge transport; thus, controlling the thermal annealing temperature and time is crucial. In general, the formation of LD capping layers immediately occurs during the casting of an organic ligand solution atop the 3D film, but with most of the ligand remaining unreacted[8,24,26,27,72,82,83,85,88]. Extending the reaction time at lower temperatures allows LD layer formation to complete with increased dimensionality. For example, for 2D perovskite layers, higher dimensionality can be achieved, typically with $n \geq 2$ (ref. 1). However, this annealing temperature control still results in a mixture of dimensional phases[1,8]. Besides, longer and lower-temperature reactions can produce more homogeneous, highly ordered LD layers[1,27], enhancing surface passivation and inhibiting ion migration. Conversely, tuning the thermal annealing temperatures can also induce the formation of different LD perovskite phases (either 0D, 1D or 2D), depending on the isomerism of the ligand[12,58]. Therefore, finding fabrication techniques and conditions that can reliably lead to the obtention of highly ordered, phase-pure and more homogeneous LD capping layers is crucial[6,18,28].

The chemical composition of the underlying 3D perovskite, such as differences in cation systems (single cation or mixed cations), has a crucial role in the formation kinetics, thermodynamics, distribution and quality of the LD perovskite layer during post-treatment. For instance, mixed-cation perovskites, which dominantly incorporate FA cations (around 80–95%) along with small quantities of other cations such as MA and/or caesium (Cs) cations, are among the most efficient and stable perovskites, with bandgaps around 1.50–1.55 eV (refs. 1,5,31,89). MA and FA are connected to halide sites through hydrogen bonds, with FA forming more hydrogen bonds than MA[5]. By contrast, Cs cations can potentially bond to halide sites through ionic bonding, which is stronger than hydrogen bonding. Thus, the in situ formation of LD perovskites on top of pure $CsPbI_3$ perovskites is more challenging and requires high processing temperatures[90]. Such thermal demands restrict their compatibility with temperature-sensitive substrates and tandem device architectures. Conversely, in situ LD formation through cation exchange on MA-based perovskites can be more readily achieved with organic ligands, owing to the smaller energy formation or weaker connection of the MA cation within the 3D perovskite matrix, enabling low-temperature processing and more homogeneous coverage[8,62,72,88]. The higher kinetic formation of the LD layer on top of 3D-$MAPbI_3$ promotes more homogeneous coverage. However, the inherent instability of 3D-$MAPbI_3$ over the long term limits this technology. Studies on 3D/LD heterostructures in pure $FAPbI_3$ are limited, possibly because of the difficulties in replacing the FA cation with bulkier organic ligands[1,7,72]. For this reason, mixed cations involving MA cations are frequently used to optimize LD layer formation (Supplementary Table 1). Beyond ligand selection, these compositional strategies can facilitate the formation of LD capping layers with varying degrees of uniformity.

In summary, 3D/LD heterostructure formation depends on ligand interactions, processing conditions and cation composition of the underlying 3D perovskite. However, how these factors ultimately influence long-term stability is still not fully understood. This uncertainty makes it necessary to examine the mechanisms of instability in more detail.

Instability mechanisms

The majority of the LD perovskites studied so far have been predominantly iodine-based[1,4,6,22–27,32,37,38,64,67,72,82,88]. However, the specific halide (I^-, Br^- or Cl^-) used to form the LD perovskites may affect the electronic properties and quality of LD perovskites themselves. Besides, the specific structure of the ligand as well as the composition of the underlying 3D perovskite can further impact the structural stability of the formed 3D/LD heterojunction[1,9,18,75,90,91]. These factors likely contribute to lab-to-lab variations in reported 3D/LD perovskite heterostructures stability, especially when using common organic ammonium ligands in solution post-treatment, highlighting the difficulty in establishing universal trends.

The degradation mechanism of 3D/LD perovskite heterostructures can stem from several factors. First, the intrinsic chemical instability of LD layers can arise from the weak chemical interactions of organic ligands. Briefly, the weak chemical interaction between adjacent ligands in the Ruddlesden–Popper phase, primarily governed by van der Waals forces, can easily disrupt the structure of the LD layer[9,58,92–94]. For this reason, the Dion–Jacobson phase can be utilized to enhance structural stability by eliminating the van der Waals gap. Another strategy involves strengthening the interlayer between ligands in the Ruddlesden–Popper phase; for example, modifying the alkylammonium 2-(methylthio)ethylammonium ligand to include an additional thiol group can simultaneously promote sulfur–sulfur interactions and van der Waals interactions[95]. Besides, using conjugative monovalent ligands can enhance the organic ligand interlayer through π–π interactions between the ligands, enhancing the structural properties of the Ruddlesden–Popper phase[17,57,96]. Second, relatively weak bonding between the organic ligands and the inorganic octahedral sheets, owing to a smaller number of hydrogen bonds, can lead to the migration of the ligands from the inorganic network[5,6,9,20,29,52,54,55]. Several attempts have been made to use multivalent ligands such as guanidinium and amidinium, which provide a greater number of hydrogen bonds with higher bond strength, thus increasing the rigidity of the LD structures[5,29,38,54]. Besides, the coordination bonding strength between the organic ligand and the inorganic sheets can be enhanced using additional functional group(s)[20,29,52,78,97,98]. The bond strength is affected by molecular symmetry and ligand dipole, which can be theoretically calculated using density functional theory. For instance, the different isomers of the aminomethyl piperidinium-based ligand can exhibit varying hydrogen bonding strengths with halide sites[99]. Similarly, amidino-pyridine ligands with multiple amine sites have been recently reported to offer stronger structural stability compared with monovalent PEA ligands under different stressors[29]. Besides, the bidentate 2-thiopheneethylammonium ligand, which introduces additional π-coordination through the thienyl ring, can also enhance the structural rigidity of the LD layer compared with regular PEA ligands[100].

In general, these weak intrinsic chemical interactions can disrupt the LD layer by forming PbI_2, metallic lead (Pb^0) and vacancy defects on the 3D perovskite surface under heat and light stresses[38,101–103] (Fig. 4a). Subsequently, a redox reaction occurs (equation (1)), driven by light and high temperatures, promoting formation of volatile Pb^0 and I_2 (refs. 101–103). In addition, ammonium ligands with low acid dissociation constant (pK_a) values (primary amines such as PEA and BA) have a tendency for deprotonation[38,102,103]. The resulting amines degrade the perovskite lattice by reacting with the A-site cations, such as FA, following the reaction, which can change the chemical structure of the ligand and detrimentally affect its passivation role[103]. The resulting NH_3 can coordinate with the 3D perovskite layer and dissolve it (equation (2)), accelerating the degradation process[38,103]. In addition, the deprotonated neutral amines can react with residual PbI_2 species through a redox reaction (equation (3))[38,102,103]. This decomposition can further lead to the formation of metallic Pb^0 and the sublimation of volatile HI or I_2, as well as to the generation of vacancy defects, which facilitate iodide migration to the anode[38,101–105] (Fig. 4b). Altogether, this process can cause device degradation and catastrophic failure.

$$PbI_2 \rightleftharpoons Pb^0 + I_2, \tag{1}$$

$$R-CH_2-NH_2 + FA^+ \rightleftharpoons R-CH=NH-NH_2^+ + NH_3 \tag{2}$$

$$R-CH_2-NH_2 + PbI_2 \rightleftharpoons R-CH=NH + Pb^0 + 2HI \tag{3}$$

As previously discussed, the variation in the kinetic formation of 1D or 2D perovskites results in uncontrolled LD perovskite formation, leading to excess unreacted ligands, increased interfacial energy mismatch and structural disorder. This issue affects charge transport, electric field distribution and chemical stability at the 3D/LD interface[4,15,20,25,35,38]. Initially, the excess organic ligand may not substantially impact device performance. However, after the organic ligands from the LD layer penetrate the 3D crystal, they can affect the chemical and physical characteristics of the 3D/LD heterostructure, altering energetic level alignment, morphology and the distribution of chemical species across the interface[15,20,38]. A washing step is occasionally necessary to remove resistive unbonded organic ligands; however, washing does not fully resolve these issues because it may result in additional chemical dissolution during the washing process, introducing further uncertainty[1,29,34,106]. Proper solvent washing engineering can be applied; however, studies on this approach remain limited. These findings raise concerns regarding the longevity of the 3D/LD heterostructure and the reported variation in the stability results of this method. To address these concerns, various strategies have been proposed, including designing suitable ligands, introducing more flexible connections between ligands and the inorganic network, such as through perovskitoid structures, replacing weak van der Waals interactions with stronger ionic or hydrogen bonding in the Dion–Jacobson phase, increasing the number and the strength of the hydrogen bonds between the organic ligands and the inorganic sheets, and fabricating engineered materials that can produce highly stable LD capping layers with excellent photo- and electronic properties.

As discussed earlier, the quality of the LD layer (including the coverage and phase purity) determines the overall stability and performance of the 3D/LD perovskites. In addition, the chemical composition of the underlying 3D perovskite substantially influences its intrinsic stability, especially under elevated temperatures and light[5,6,107]. For example, all-inorganic 3D perovskites have better thermal resilience at high temperatures[90]. By contrast, $MAPbI_3$ is highly susceptible to thermal stress owing to the high volatility of the MA cation[5]. $FAPbI_3$ perovskites offer improved thermal and photo-stability, particularly when combined with mixed A-site cations such as formamidinium cations or FAMACs (mixed-cation system containing formamidinium, methylammonium and caesium cations)[1,5,108]. These compositional

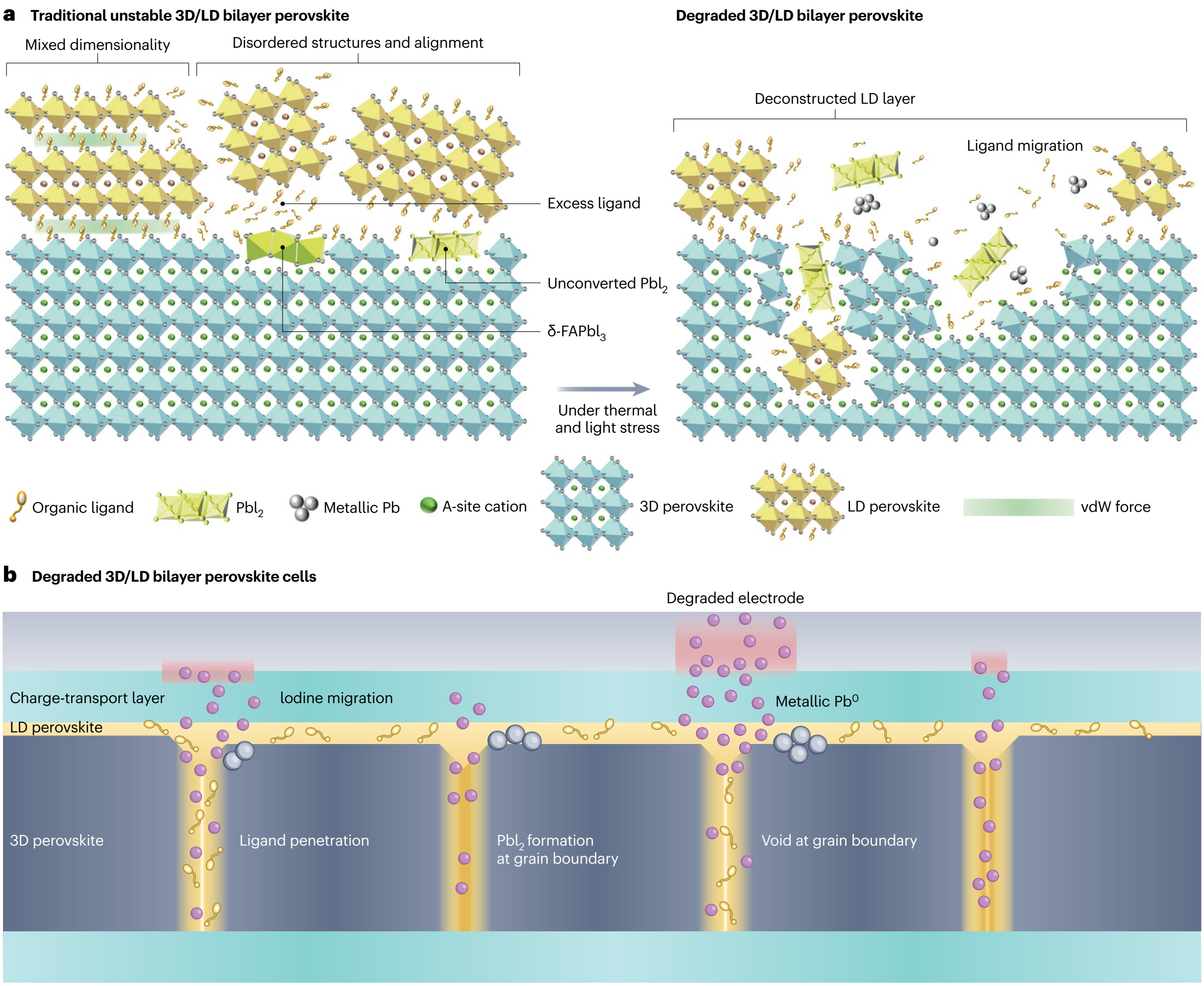


Fig. 4 | 3D/LD interface deconstruction. a, Interfacial 2D perovskite layer deconstruction under heat or illumination. **b**, 3D/low-dimensional (LD) bilayer degradation through grain boundary-mediated void formation. These sites can facilitate redox reactions, leading to the formation of excess PbI_2, metallic Pb and ammonium-containing liquids that can dissolve the underlying perovskite layer. vdW, van der Waals.

strategies can contribute to enhancing the structural and chemical stability of the resulting heterojunction.

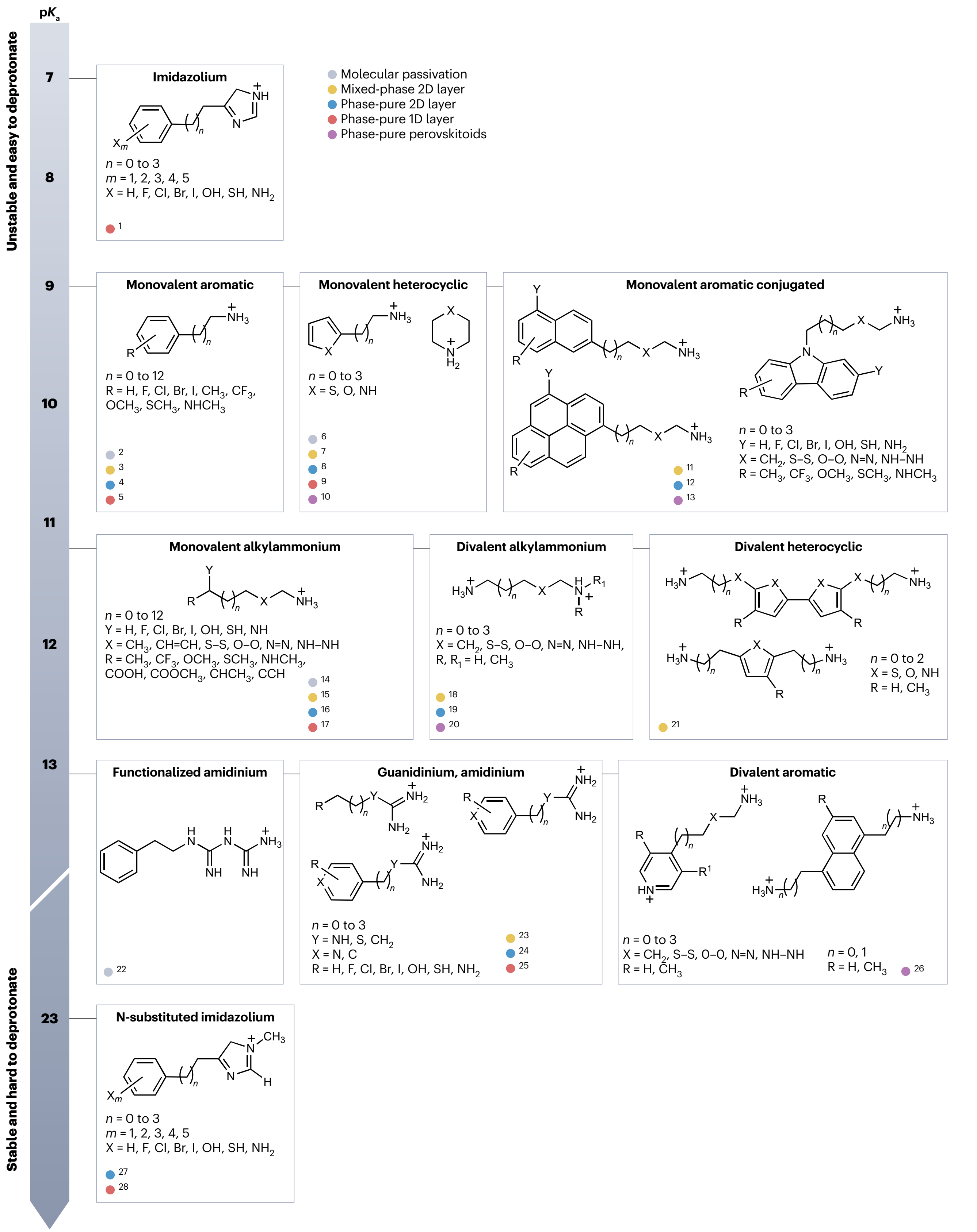
Ultimately, the inherent instability of LD perovskite capping layers arises from the properties of the organic ligands, quality of the LD capping layer and the composition of the perovskites. Therefore, it is essential to construct high-quality LD capping layers with desired optoelectronic properties through ligand design and fabrication engineering.

Engineering stable low-dimensional capping layers

Organic ligands have a crucial role in stabilizing 3D/LD heterostructures by binding to the perovskite surface and reducing potential degradation pathways. However, their chemical instability can lead to the detachment of the ligand from the perovskite layer and the decomposition of the remaining film under stress, exacerbating perovskite instability. Developing more robust ligands or modifying existing ones is essential to enhance their durability and performance. One strategy is to enhance the stability of the organic–inorganic interface. Another strategy involves optimizing the synthesis process of the 3D/LD heterostructure to achieve a homogeneous and highly phase-pure LD capping layer on the 3D perovskite surface.

Ligand chemistry and design

Substantial research has been dedicated to exploring novel ligands to form more effective 2D and 1D perovskite passivation layers. However,



pKa
Unstable and easy to deprotonate
Stable and hard to deprotonate
7
8
9
10
11
12
13
23
Molecular passivation
Mixed-phase 2D layer
Phase-pure 2D layer
Phase-pure 1D layer
Phase-pure perovskitoids
Imidazolium
n = 0 to 3
m = 1, 2, 3, 4, 5
X = H, F, Cl, Br, I, OH, SH, NH2
1
Monovalent aromatic
n = 0 to 12
R = H, F, Cl, Br, I, CH3, CF3, OCH3, SCH3, NHCH3
2
3
4
5
Monovalent heterocyclic
n = 0 to 3
X = S, O, NH
6
7
8
9
10
Monovalent aromatic conjugated
n = 0 to 3
Y = H, F, Cl, Br, I, OH, SH, NH2
X = CH2, S–S, O–O, N=N, NH–NH
R = CH3, CF3, OCH3, SCH3, NHCH3
11
12
13
Monovalent alkylammonium
n = 0 to 12
Y = H, F, Cl, Br, I, OH, SH, NH
X = CH3, CH=CH, S–S, O–O, N=N, NH–NH
R = CH3, CF3, OCH3, SCH3, NHCH3, COOH, COOCH3, CHCH3, CCH
14
15
16
17
Divalent alkylammonium
n = 0 to 3
X = CH2, S–S, O–O, N=N, NH–NH,
R, R1 = H, CH3
18
19
20
Divalent heterocyclic
n = 0 to 2
X = S, O, NH
R = H, CH3
21
Functionalized amidinium
22
Guanidinium, amidinium
n = 0 to 3
Y = NH, S, CH2
X = N, C
R = H, F, Cl, Br, I, OH, SH, NH2
23
24
25
Divalent aromatic
n = 0 to 3
X = CH2, S–S, O–O, N=N, NH–NH
R = H, CH3
n = 0, 1
R = H, CH3
26
N-substituted imidazolium
n = 0 to 3
m = 1, 2, 3, 4, 5
X = H, F, Cl, Br, I, OH, SH, NH2
27
28

Fig. 5 | Ligand chemistry design and classification. Classification of ligands based on their chemical structure and on the acid dissociation constant (pK_a) of their ammonium cation. Examples for each ligand classes are selected from publications that emphasize phase purity and the dimensionality of the resulting LD perovskite and perovskitoids layers. 1, refs. 123,124; 2, refs. 140–144; 3, refs. 8,16,24,25,72,91,108,145–152; 4, refs. 19,20,36,67,112,135,153,154; 5, refs. 13,155; 6, ref. 113; 7, refs. 14,156,157; 8, ref. 158; 9, refs. 11,159–161; 10, ref. 45; 11, ref. 17; 12, ref. 97; 13, ref. 48; 14, refs. 162–168; 15, refs. 1,7,35,64,65,82,85,109,127, 169–172; 16, refs. 15,18,21,66,173–176; 17, refs. 177–181; 18, ref. 9; 19, refs. 48,64, 116–187; 20, refs. 46,47; 21, refs. 188–191; 22, ref. 192; 23, refs. 193,194; 24, refs. 10,118,195; 25, ref. 196; 26, ref. 12; 27, ref. 121; 28, refs. 120,122.

few reports focus on the homogeneity of the capping layer, which is crucial especially for large-sized perovskite modules[4,27,70,85,109]. In general, the formation of LD perovskites is influenced by six factors related to ligands, such as their charge, shape, size, solubility, hydrogen bonding and reactivity[6,56,98]. Converting the head group to FA or guanidinium moieties, which have high pK_a values, can increase the stability of the 3D/LD heterojunction[1,6,7,26,29,38,85,103]. The type of cations used for surface treatment affects the extent of these changes: although traditional cations with short carbon chains are commonly used, those with long carbon chains provide more stable interfaces[1,4,27,34]. We classify each type of ligand molecules in Fig. 5 according to their chemical structure and the pK_a values of the ammonium cations.

Most ligands with ammonium head groups can passivate or react with A-site defects even at low concentration[6,110–113]. When the concentration of the ligand is increased to a critical value, LD perovskites start forming on the surface of the 3D perovskite in situ. These ligands can form either LD perovskites or perovskitoids depending on the organic ligand arrangements within the inorganic octahedral networks, as discussed previously. The LD phase structure can be influenced by variations in the electrostatic potential of the isomeric ligands and the steric effects of intermolecular forces[11–14]. Another factor influencing the ultimate dimensionality of the phase (0D, 1D or 2D) is the ratio between the cations and anions of the ligands, as well as the type of anion, owing to differences in the formation energy between each LD phase[11,12,73]. Ionic liquids containing N-substituted imidazolium ligands tend to form 1D phases rather than 2D phases owing to a favourable formation energy[75,114]. However, the fundamental formation mechanism of the different LD phases still remains unclear, and more systematic studies are needed.

Depending on the size and reactivity of the ligands, the 3D/LD heterojunction either forms at the outermost surface of the 3D perovskite, or the ligand may penetrate deeper into the grain boundaries[57]. Hence, we classify common ammonium ligands by their size, functionality and pK_a values. Smaller monovalent and divalent ligands can penetrate deeper into the bulk and grain boundaries of the 3D perovskite, leading to 2D layer with mixed dimensionality ($n = 1$–5)[78,81,111]. By varying the processing conditions such as the solvent and annealing temperature, ligands can be limited at the surface, with controlled dimensionality[115]. For example, tuning the polarity of the solvent can help to modulate the penetration depth of the organic ligands. Solvents with higher polarity, such as 2-propanol, can help the ligands to penetrate deeper into the 3D perovskite film, whereas less polar solvents, such as chlorobenzene, restrict the ligands to the very surface of the 3D perovskite film. Moreover, depending on the activation energy for 2D perovskite formation, post-annealing at higher temperatures tends to produce the most thermodynamically stable 2D phase, whereas post-annealing at lower temperatures can favour the growth of alternative, less stable 2D phases[1].

Monovalent heterocyclic and aromatic ammonium ligands struggle to penetrate the 3D perovskite structure owing to their bulky size and the ability of aromatic rings to arrange into π–π stacked structures. This limited penetration often results in 2D capping layers with higher dimensionality ($n \geq 3$)[8,17,116]. By contrast, when aromatic ligands are extended into larger conjugated systems, steric effects further restrict their insertion, leading to the formation of lower-dimensionality 2D layers ($n = 1$ or 2)[16,17,97]. Although only a single phase is typically observed in 1D perovskites owing to constraints in crystal structure arrangement[11–13], the formation of LD perovskitoids is more complex and depends on the ligand to PbI_2 ratio as well as the shape and functionality of the ligand itself. As a result, no systematic trends have been established so far.

The stability of the formed LD perovskite also correlates with how well the ligand spacers can stay in place without migration and degradation[38]. Lewis bases are the main cause behind the decomposition of ammonium ligands in the perovskite film, resulting in deprotonation of the ammonium group, loss of positive charge and ligand migration[6,38,103]. The deprotonation of ammonium ligands can be mediated by weak Lewis bases such as moisture-containing and oxygen-containing compounds that can assist in removal of a proton from the ammonium group under operational conditions or exposure to air[38]. Moreover, light and thermal stress can assist in oxidation of iodide into iodine, leading to the formation of triiodide (I_3^-) species that can act as a weak Lewis base in deprotonating the ammonium ligands[81]:

$$H_2O + R-NH_3^+ \rightarrow H_3O^+ + R-NH_2 \quad (4)$$

$$I_3^- + R-NH_3^+ \rightarrow I_2 + HI + R-NH_2 \quad (5)$$

In both cases, highly mobile amines, formed inside the perovskite, can migrate along the grain boundaries and participate in further acid–base reactions that can cascade into deprotonation and evaporation of the highly volatile methylamine as a gas. To prevent deprotonation and improve LD perovskites stability, ligand spacers can be converted into moieties with higher pK_a values[103], with guanidinium, amidinium and N-substituted imidazolium groups being the most viable[29,54,65,75,114,117–119]. Primary ammonium groups have pK_a values between 9 and 11, making them susceptible to deprotonation by strong Lewis bases with high pK_b values[6,30,31,34,36,38,103,111]. Guanidinium and amidinium moieties have pK_a values above 12–13, making them more stable against deprotonation and decomposition[54,65,75,114,117–119]. In this regard, N-substituted imidazolium groups provide ligands with the best stability against deprotonation, as the nitrogen atom does not carry a proton that could be removed. The most acidic proton in the imidazole ring is bound to a carbon atom and has a pK_a value above 23. Owing to this effect, ionic liquids with N-substituted imidazolium groups can form excellent phase-pure 1D or 2D perovskites[120–122]. Interestingly, imidazolium with no N-substitution has the lowest pK_a value among LD ligands (close to 7) and is therefore expected to participate in Lewis base-assisted degradation more readily than other ligands[123].

The structure, functionality and steric availability of the molecules are critical for the formation of 1D and 2D perovskites and must be carefully considered when designing ligands[6,53,56,98]. Sterically unavailable

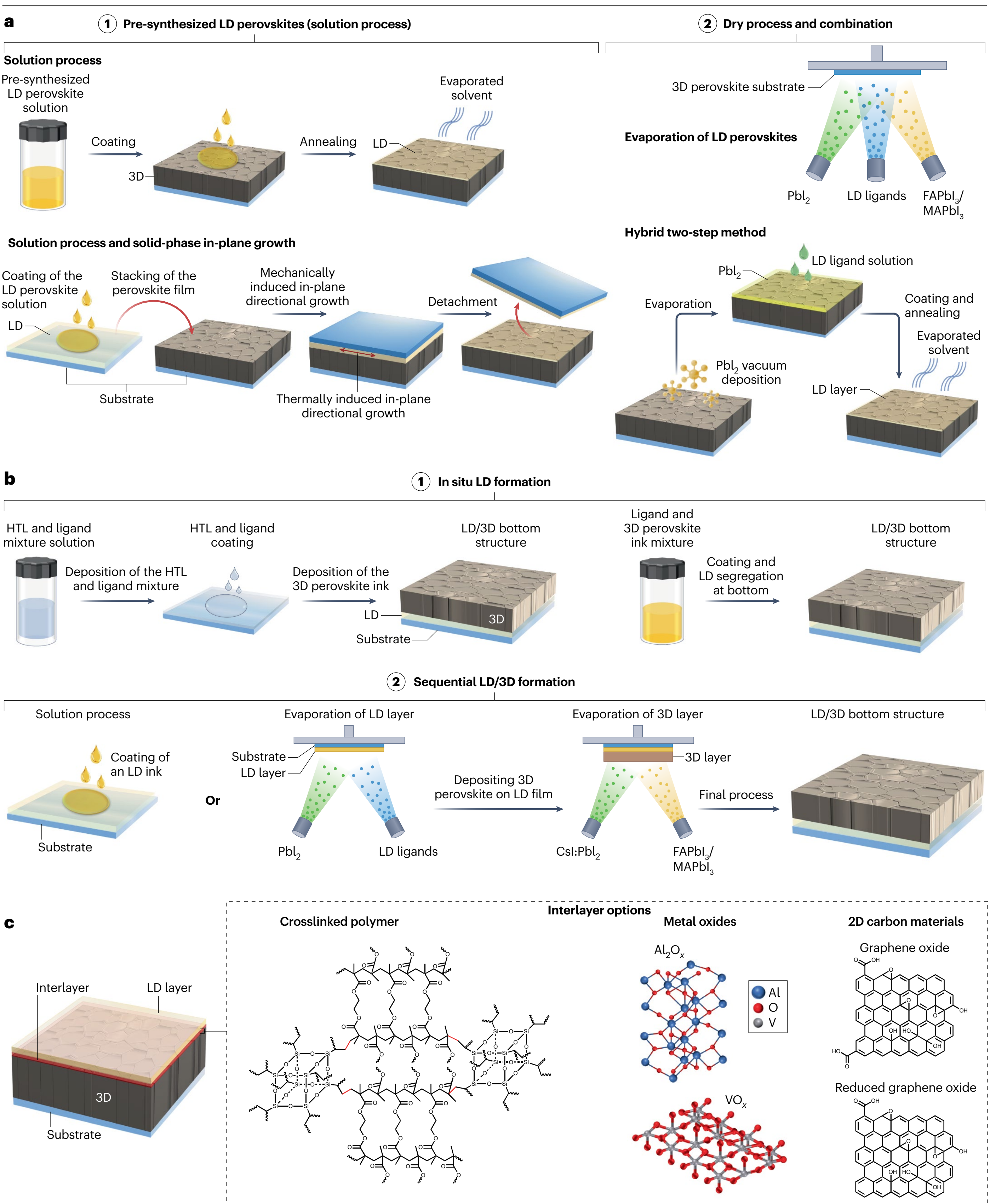

a
1 Pre-synthesized LD perovskites (solution process)
2 Dry process and combination
Solution process
Pre-synthesized LD perovskite solution
Coating
3D
Annealing
Evaporated solvent
LD
3D perovskite substrate
Evaporation of LD perovskites
PbI2
LD ligands
FAPbI3/MAPbI3
Solution process and solid-phase in-plane growth
Coating of the LD perovskite solution
LD
Stacking of the perovskite film
Mechanically induced in-plane directional growth
Detachment
Substrate
Thermally induced in-plane directional growth
Hybrid two-step method
LD ligand solution
PbI2
Evaporation
Coating and annealing
Evaporated solvent
PbI2 vacuum deposition
LD layer
b
1 In situ LD formation
HTL and ligand mixture solution
HTL and ligand coating
LD/3D bottom structure
Ligand and 3D perovskite ink mixture
LD/3D bottom structure
Deposition of the HTL and ligand mixture
Deposition of the 3D perovskite ink
LD
3D
Substrate
Coating and LD segregation at bottom
2 Sequential LD/3D formation
Solution process
Evaporation of LD layer
Evaporation of 3D layer
LD/3D bottom structure
Coating of an LD ink
Substrate
Or
Substrate
LD layer
PbI2
LD ligands
Depositing 3D perovskite on LD film
3D layer
CsI:PbI2
FAPbI3/MAPbI3
Final process
c
Interlayer options
Crosslinked polymer
Metal oxides
2D carbon materials
Interlayer
LD layer
3D
Substrate
Al2Ox
Al
O
V
VOx
Graphene oxide
Reduced graphene oxide

Fig. 6 | Phase-pure LD layer fabrication and engineering. a, Phase-pure and homogeneous low-dimensional (LD) perovskites obtained through fabrication engineering, using pre-synthesized LD perovskites via solution processing and dry vacuum-assisted methods. **b**, Fabrication engineering of an LD capping layer at the bottom contact. **c**, Method to enhance 3D/LD perovskite heterostructure stability through interlayer insertion. HTL, hole transport layer.

monovalent and divalent cations cannot form LD perovskites owing to the orientation of the two ammonium head groups or hindered cation heads that prevent them from occupying A-site cation space[6,111,115]. For example, in benzene-1,2-diethanammonium, the two ammonium heads are located on the same side, making it impossible to coordinate with two different A-sites[115]. Similarly, molecules with multiple functional groups that can strongly interact with the 3D perovskite surface, such as phenformin hydrochloride, cannot form LD perovskites because of the favourable interactions between the perovskite surface defects and these functional groups[124].

The Ruddlesden–Popper phase benefits from easy processing and a wide variety of highly soluble ligands. By contrast, the Dion–Jacobson phase is constrained by the required use of divalent amine ligands, which dissolve poorly in polar solvents owing to the strong propensity of their $-NH_3^+$ groups to form hydrogen bonds with halide ions in solution. This difference makes the Ruddlesden–Popper phase attractive for simpler synthesis and scalability. Besides, interdigitating monovalent ligands can induce better self-assembly, ultimately forming highly oriented and crystalline Ruddlesden–Popper layers. Despite these processing difficulties and the lower degree of conformation freedom inherent to divalent ligands, the Dion–Jacobson phase offers relatively better charge transfer owing to shorter distance between the two inorganic layers, making it preferable for specific high-performance applications. To address the aforementioned challenges, researchers have devised novel Dion–Jacobson spacers or co-solvent systems (typically with high Lewis basicity, such as dimethyl sulfoxide, *N*-methyl-2-pyrrolidone or *N*,*N*′-dimethylpropyleneurea). For instance, the ether-bridged diammonium 2,2-(ethylenedioxy)bis(ethylammonium) tends to form very stable hydrogen bond networks that 'lock in' an $n = 1$ Dion–Jacobson phase without mixed domains[125]. Similarly, dipolar diamines such as dimethyl adipimidate^{2+} can moderate nucleation by forming hydrogen bonds in precursor solutions, which help to produce uniformly sized colloids and spontaneously yield phase-pure 2D Dion–Jacobson layers[116]. However, excessive hydrogen bonding or mismatched ligand geometry can distort the lattice and raise defect densities. Thus, optimizing Dion–Jacobson ligands requires careful management of the trade-off between hydrogen bonding strength, which should be sufficient to guide phase formation, and excessive interactions that risk locking the lattice into an energetically unfavourable configuration. Furthermore, to tackle the solubility issue, researchers have turned to compiled solvent systems and multicomponent processes[29,35,57,63,65,77,87]. Despite these disadvantages, divalent ligands, such as aromatic and aliphatic diammonium cations, can form LD perovskitoids with various dimensionalities and lattice connectivity depending on the shape and size of the ligand itself[12], which makes them an excellent choice for fabrication of novel LD perovskitoids.

As another option, asymmetric diammonium molecules with different substituents on each end can modulate packing and electronic coupling. For example, asymmetric 1,3-propane diammonium (with one tertiary amine) was used to form a Dion–Jacobson phase 'head-to-tail' capping layer, providing better band alignments with symmetric spacers[9,126]. In general, asymmetry can break unfavourable ligand–ligand interactions and introduce an internal dipole, but it complicates synthesis and often reduces solubility. Despite the ligand innovation, aromatic and cyclic spacers offer different opportunities. Rigid conjugated diamines (benzene-based or naphthalene-based) can increase framework stiffness and introduce π–electron interactions. However, aromatic diammonium often suffers from π–π stacking: molecules that pack face-to-face tend to collapse the Dion–Jacobson lattice into different LD (0D or 1D) phases[68]. Only benzene-derived diammonium ligands with weak intermolecular π–π interactions form stable 2D Dion–Jacobson phases; strong π–π stacking leads to phase decomposition. Designing asymmetrical or linked aromatics (with small dipoles or steric hindrance) alleviates this issue and helps to maintain the Dion–Jacobson structure. Importantly, short π-conjugated spacers can greatly improve charge transport. For example, 3-aminobenzylammonium or 4-aminobenzylammonium, short benzene-terminated molecules with high dipole moments, produce 2D Dion–Jacobson perovskites with extremely low exciton binding (~84 meV) and fast charge transfer[92]. Cyclic aliphatic ligands, such as 4-aminomethylpiperidinium, similarly add rigidity. They produce greater distortion but also higher stability than linear chains[126]. However, the use of rigid aromatic or cyclic ligands is often limited by their poor solubility in polar solvents and their tendency to induce lattice strain through enhanced steric bulk and polarizability.

Fabrication of phase-pure films

Several studies have addressed instability in 3D/LD heterostructures, suggesting that 2D perovskites with a uniform *n* value are preferable as passivation layers to ensure efficient and stable photovoltaic devices. These passivation layers can be obtained using solution processing by directly depositing pre-synthesized 1D or 2D crystals or through dry methods such as vacuum evaporation and hybrid two-step deposition (Fig. 6a). For instance, by carefully controlling temperature and pressure, a solid-state deposition technique using pre-synthesized, phase-pure 2D-BA_2PbI_4 ($n = 1$) films has been used to enable in-plane stacking on top of the 3D perovskite film, resulting in a 2D film with controlled thickness, improved quality and homogeneous planar coverage. This method also prevents the formation of unintended phases between the 2D and 3D layers, which is beneficial for high-quality junctions with minimized interfacial recombination and enhanced built-in potential. As a result, this strategy yields efficient and stable 2D/3D PSCs, achieving a maximum power conversion efficiency (PCE) of 24.8% and retaining 94% of their initial efficiency after over 1,000 h of damp-heat and continuous 1-sun illumination testing.

As another approach, pre-synthesized phase-pure LD inks can be coated onto the 3D film using solvent engineering to minimize the 3D perovskite dissolution underneath. This ink can be deposited using several techniques, such as spin-coating, blade-coating or slot-die-coating, to ensure a conformal 2D layer on top of the 3D film. For example, polar aprotic solvents with high dielectric constants and donor strengths can be used to dissolve pre-synthesized 2D perovskites with a specific *n* value, enabling deterministic fabrication of 3D/2D perovskite heterostructure stacks and the formation of uniform 2D layers with controlled dimensionality[18]. Adjusting the solvent dielectric constants and donor strengths both allow the dissolution of the pre-synthesized 2D perovskites, allowing the solution-processed coating of the 2D perovskite layer, and prevent the dissolution of the 3D perovskite underneath. Using this approach, a device with a 2D-$BA_2MAPb_2I_7$ (n = 3) capping layer showed a maximum PCE of around 24.5% with almost no PCE loss under 1 sun and at 60 °C for around 2,000 h. However, the difficulty to synthesize phase-pure 2D perovskite with higher dimensionality ($n \geq 3$) remains a concern and needs more attention before this approach can be fully realized. Besides phase purity concerns, the scalability of the aforementioned techniques remains uncertain, particularly for 2D perovskites obtained using large-space or long-chain alkylamine ligands. Although these ligands offer greater stability owing to their hydrophobicity and strong steric hindrance, it is unclear whether these techniques can be broadly applied to such systems. It is noted that the direct deposition of pre-synthesized 1D and 0D perovskites is feasible through this method, but there are limited reported studies so far.

Although the aforementioned techniques have been successful in enhancing the stability of perovskite devices, their PCEs remain relatively low compared with state-of-the-art PSCs. This difference suggests that better passivation and LD quality are needed to enhance the efficiency of 2D/3D PSCs. In addition, obtaining highly phase-pure and oriented 2D perovskites remains a challenge because of various factors[28]. To obtain phase-pure 2D perovskite layers, a two-step hybrid method can be used: first, a controllable thickness of PbI_2 is deposited on pre-prepared 3D perovskites using vacuum evaporation; then, organic ligands (or a mixture of ligands and $FAPbI_3$) are spin-coated onto the surface, followed by thermal annealing. This process enabled the formation of 2D perovskites with precisely controlled phase purities and dimensions[20]. This method has been linked to an increase in device performance to 25.6% and has scale-up potential; however, the full conversion of the thin PbI_2 layer to the 2D capping layer remains to be explored. Therefore, it is critical to control the formation kinetics of the 2D structure to simultaneously achieve high efficiency and stable heterostructures. Vapour deposition has also been used to form a 2D Ruddlesden–Popper perovskite capping layer with high homogeneity and phase purity[67]. This method is also scalable while still allowing precise control over the composition and thickness of the 2D or 1D perovskites. However, as future research progresses, it is recommended that these techniques be further refined to enhance phase purity, orientation and stability, ultimately leading to more efficient and durable perovskites[3,6,20,27,28,56,87].

Discussion so far has focused only on top-contact passivation. Bottom-contact passivation has traditionally been overlooked owing to challenges associated with the solution-based fabrication process. Specifically, when forming the LD perovskite passivation layer beneath the 3D perovskite film, the solution used to deposit the 3D layer can inadvertently dissolve the underlying LD perovskite[20,80,108,127,128]. Several strategies have been introduced to solve this dissolution challenge (Fig. 6b). For example, mixing organic ligands with bottom electron-transporting or hole-transporting layers can strengthen the ligand interaction with the substrate, and while subsequently coating the 3D perovskite, a 2D perovskite layer can be formed at the bottom substrate–perovskite interface[20,127]. Another strategy involves mixing the organic ligand 2-aminoindan, which has a strong interaction with PbI_2, into the perovskite ink, resulting in pre-formed 2D perovskites[108]. These pre-crystallites 2D perovskites then sediment at the bottom substrate, inducing the formation of the 2D perovskite layer at the bottom interface. However, these strategies typically result in a random and discontinuous 2D perovskite layer at the bottom interface. To obtain more uniform and continuous 2D perovskite layers, evaporation or hybrid methods can also be used[80]. The 2D perovskite passivation layer at the bottom substrate is important not only to passivate the defects but also in guiding the crystallization of the 3D perovskite and stabilizing its formation by acting as a template for the growth of the 3D perovskite film[128]. It is noted that LD perovskite bottom passivation is still limited to 2D perovskites, as there are no reports for 0D and 1D perovskites. Overall, the formation of double-sided 3D/LD heterojunctions is desirable; however, there is still a lack of systematic research into the design of suitable organic ligands, the exploration of different LD structures and the development of scalable fabrication methods.

Interlayer engineering

Inhibiting ion diffusion between the 3D and LD perovskites and stabilizing the 3D/LD perovskite heterojunction remain a challenge if unstable ligands are used, as discussed earlier. To resolve this, crosslinked polymers have been deposited on top of 3D perovskite layers, followed by the deposition of 2D-PEA perovskite layers using a vapour-assisted two-step process to form a 3D/crosslinked polymer/2D perovskite heterostructure[36]. This interlayer effectively inhibits ion diffusion without hindering charge transport between the 3D-PEA and 2D-PEA perovskite layers. It is also important to note that to maintain the property of 3D/LD, the interlayer materials should be thin enough to avoid charge recombination and transport issues. Other types of interlayers have also been explored to minimize 3D perovskite film damage during subsequent LD deposition. Towards this goal, promising robust and uniform interlayer candidates include 2D materials that can match the crystal phase and orientation of the LD perovskite, thereby facilitating its growth over large areas and with precise controlled thickness. Examples of such 2D materials include metal oxides, other conjugated polymers and graphene and derivatives, which can be deposited using scalable techniques[129] (Fig. 6c). The interlayer strategy could potentially improve the performance and stability of 3D/LD heterostructures, opening alternative ways to simplify the deposition of pre-synthesized LD perovskites (with different structures and dimensionalities) using both scalable solution and dry vacuum methods.

Stability analysis and advances

Although there has been substantial progress in enhancing stability under operational temperature and outdoor settings, the anticipated performance of 3D/LD-based PSCs must also be gauged directly over years to decades[1,4,18,20,29,59,60,90]. Depending on the LD ligand and its quality, 3D/LD interfaces begin to degrade at ~60 °C or above; however, outdoor solar module surfaces can reach an operational temperature above 70 °C for prolonged periods[130,131]. Therefore, it is vital to examine their stability at higher temperatures, for example, at ≥85 °C, during accelerated tests[20,54,90,111]. Cation exchange between the LD and 3D phases also creates new, unstable phases. To assess the resilience of device stacks to cycles and external stresses, outdoor field testing is essential[130]. To this end, developing an accelerated model to describe the perovskite response to degradation is becoming increasingly crucial. Such a model should capture perovskite-specific phenomena, distinct from established silicon photovoltaics, particularly concerning ion migration and performance reversibility[107].

Conventionally, linear extrapolation has been widely used to assess solar cell degradation. Devices often exhibit an initial fast exponential decay phase 'burn-in' (or even an initial performance improvement) followed by a stabilized region[132]. However, this behaviour is not observed in all devices; further degradation and even potential recovering under dark are often observed[132,133]. Alternatively, degradation can be artificially accelerated by applying an external stressor such as elevated temperature, illumination or mechanical stress. In organic photovoltaics, degradation behaviour can be modelled using a biexponential function, followed by an Arrhenius-type behaviour[107,133] (Box 1). Recently, similar trends have been observed in perovskite photovoltaics[90,107]. We can then assess the effects of the LD capping layer on device longevity, based on stability data from published studies on encapsulated cells under continuous illumination. We use the Arrhenius factor to standardize measurements at the same temperature (35 °C) owing to temperature differences in each study[6,90,91,107]. This reference temperature reflects a typical outdoor operating environment where sun irradiation often raises temperatures above standard room levels[107,130]. It is important to note that temperature can vary depending on location and testing time[131,134].

The operational lifetime of devices with LD capping layers is estimated to be several months at this set temperature (35 °C), with some instances lasting up to several years (Fig. 7). Upon depositing the LD capping layer, the T_{80} lifetime (defined as the time needed for the PCE to drop to 80% of its initial PCE) of the perovskite devices can improve by a factor of two and, in some instances, by more than 10-fold compared with reference cells that have typical T_{80} lifetimes of less than 1,000 h (refs. 18,20,21,36,48,90,132). In some cases, the presence of the capping layer alters the degradation pattern, transitioning from a burn-in behaviour to a non-monotonic trend, in which the stabilized PCE reaches a higher value[6]. However, this improvement may be overestimated because of the emphasis on demonstrating enhanced stability compared with control cells. Control 3D-based PSCs with different compositions may exhibit varying lifetimes, with some degrading within a month and others remaining operational for several months, which may influence how the stability advantage of 3D/LD devices is perceived[1,18]. In perovskites utilizing inorganic $CsPbI_3$ as opposed to $FAPbI_3$ and $MAPbI_3$, the A-site bond strength is higher, resulting in thermal stability at temperatures exceeding 100 °C. By contrast, FA-based cells remain stable below 95 °C, and MA-based cells remain stable below 60 °C (refs. 5,90,91). By comparing different LD capping distributions, specifically planar homogeneous thin layers and discontinuous layers (infiltrated at the grain boundaries and on the surface), it was observed that planar homogeneous layers typically have longer lifetimes[18,36,135]. It is also important to note that the choice of ligands and dimensionality tuning of LD affect stability results. In general, ligands with higher pK_a and lower dimensionality tend to be more stable[103]. These trends correlate with previously discussed engineering strategies aimed at combining uniform coverage with stronger interfacial interactions.

In general, Dion–Jacobson perovskites deliver superior operational stability compared with Ruddlesden–Popper perovskites[38,93,136]. This difference stems from their unique structural features, such as the presence of divalent diammonium cations in the Dion–Jacobson phase, which form hydrogen bonds with adjacent Pb–I slabs on both sides, eliminating the weak van der Waals gaps present in Ruddlesden–Popper structures[9,29,38,93,94,99,126]. The resulting tighter interlayer coupling enhances lattice rigidity, effectively suppresses ion migration and makes Dion–Jacobson frameworks less vulnerable to moisture penetration and thermal stress[9,93,94,136]. Dion–Jacobson 2D perovskite passivation has been successfully used to enhance the operational stability of perovskite devices compared with the Ruddlesden–Popper phase[9,38,68,92–94,99,126]. For example, Dion–Jacobson-based 2D/3D PSCs have a remarkable operational stability with T_{80} lifetime under extreme conditions (two-suns illumination at 85 °C) for more than 560 h, whereas Ruddlesden–Popper 2D/3D PSCs undergo degradation 10 times faster under identical conditions[38].

Box 1 | Modelling of degradation behaviour

The change in power conversion efficiency (PCE) mirrors the decay rate function and is represented by the rate constant k:

$$\mathrm{PCE}(t) = f(k_1, k_2, k_3, \ldots, k_x) + C \tag{6}$$

Here, each k represents different decay modes reflected in the material (ion migration, segregation and diffusion), which can be expressed as:

$$k(T) = A\exp\left(\frac{-E_a}{k_B T}\right) \tag{7}$$

with $k(T)$ being the temperature-dependent degradation rate, A a constant representing the rate without thermal activation, E_a the activation energy, k_B the Boltzmann constant and T the temperature. The value of E_a is correlated with the perovskite composition. For example, $MAPbI_3$-based perovskite solar cells (PSCs) have an E_a between 0.4 eV and 0.6 eV, whereas the E_a for $FAPbI_3$-based PSCs ranges between 0.5 eV and 0.7 eV (refs. 90,182–184). These values indicate that although E_a depends on composition, the E_a of different PSCs remains comparable within the typical range of perovskite formulations.

By rearranging the mathematical operation from equation (6) the value of $-E_a$ can be derived from the slope between $\partial\ln(k(T))$ and $1/k_B T$ (ref. 90).

As the decay rate depends strongly on the temperature, the model can be further developed to calculate the acceleration factor (AF)[180]:

$$\mathrm{AF} = \frac{k_{T1}}{k_{T2}} = \exp\left(\frac{E_a}{k_B}, \left[\frac{1}{T_2} - \frac{1}{T_1}\right]\right) \tag{8}$$

which is the ratio between the degradation rate at two different temperatures. This model provides an effective tool for estimating the constant temperature at which degradation is equivalent to that observed in accelerated laboratory tests, enabling reliable extrapolation from controlled conditions to real-world performance.

However, several studies have reported impressive performance from Ruddlesden–Popper 2D/3D PSCs, achieved through fabrication engineering aimed at producing phase-pure LD layer[4,18,20,21,29,48] (Supplementary Table 1). For example, Ruddlesden–Popper films built with BA spacers retain 99% of their initial PCE after 2,000 h of continuous one-sun illumination at 55 °C and 65% relative humidity[18]. Under the same conditions, a conventional 3D device falls to 75% of its starting PCE in just 100 h. Conversely, LD perovskites that exhibit continuous coverage but contain mixed phases, or those with discontinuous coverage despite being phase-pure, tend to show only moderate operational lifetimes, with T_{80} values spanning several months to a few years. By contrast, samples with mixed-phase or discontinuous 2D perovskites have moderate-to-low lifetimes.

Our investigation reveals that phase-pure LD layers improve stability not only by eliminating energetic disorder and reducing charge-transfer losses but also by providing structural homogeneity, where uniformly protected inorganic slabs suppress phase segregation

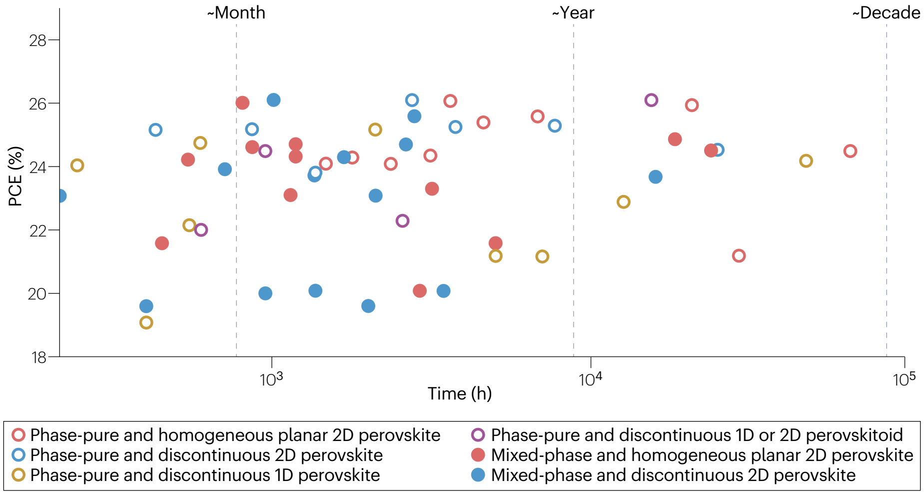


Fig. 7 | Reported stability for different types of low-dimensional capping layers and their maximum power conversion efficiencies. The stability is measured under 1 sun and at elevated temperatures. The time needed for the power conversion efficiency (PCE) to drop to 80% of its initial PCE (T_{80}) lifetime is estimated using the Arrhenius model at a set temperature of 35 °C (refs. 107,133). The data are summarized in Supplementary Table 1.

and defect formation[20,28]. The relatively modest PCE observed is instead attributed to band alignment and charge-transport limitations at the 2D/3D interface, rather than to the intrinsic quality of the LD layer[20,28]. These results highlight the combined importance of phase purity and uniformity in passivating and inhibiting ion migration. It is instructive to consider the underlying thermodynamic properties to better understand the intrinsic factors governing such stability differences between Ruddlesden–Popper and Dion–Jacobson phases. Ultimately, both Ruddlesden–Popper and Dion–Jacobson 2D perovskites still have substantial room for improvement, and targeted phase engineering will be essential to fully harness their complementary stability advantages.

LD perovskitoids typically exhibit different trends. Unlike Ruddlesden–Popper and Dion–Jacobson perovskites, which form continuous dipolar layers that passivate surface defects and create energetic barriers to ion migration, 1D perovskitoids frequently use spacers that pack in edge-sharing geometries or with alternating orientations. This arrangement causes intrinsic and induced dipoles to partially cancel, reducing the net interfacial dipole and undermining the suppression of halide migration[137]. Moreover, LD perovskites or perovskitoids often incorporate polyaromatic spacers such as pyrene derivatives that can harvest and trap triplet excitons. These long-lived triplet states readily transfer energy to the oxygen present in the surrounding moisture or solvent, generating reactive singlet oxygen species that attack both the inorganic lattice and the organic spacer. Unlike their 2D counterparts, whose thick and uniform hydrophobic layers act as effective barriers to oxygen diffusion, 1D chains present more exposed surface area per spacer and thus accelerate photo-oxidative degradation[138]. Considering those factors, the long-term stability 1D perovskites or perovskitoids rely on triplet management strategies such as heavy-atom substitution or incorporation of molecular triplet quenchers, which can suppress long-lived excited states and inhibit singlet-oxygen generation[138,139]. Additionally, precise band-edge tuning using compositional or interfacial dipole engineering can prevent carrier accumulation at heterojunctions[38,126].

When selecting an LD perovskite, it is essential to balance the intrinsic thermodynamic stability of the material with its extrinsic resistance to photothermal degradation and ion migration. Beyond selecting specific phases and structures for LD perovskite and perovskitoid frameworks, molecular engineering strategies such as tuning the dipole moment of spacer cations can enhance interlayer interactions and reinforce structural rigidity against environmental stressors. Moreover, managing excited-state dynamics by mitigating long-lived triplets presents an effective route to suppress singlet oxygen formation, a key contributor to photo-oxidative degradation. Together, these design principles position tailored LD perovskites as highly promising materials for achieving long-term operational stability in photovoltaic devices under real-world conditions.

In summary, the stability of LD-based perovskite devices is governed by a complex interplay of thermodynamic, structural and interfacial factors, with phase purity, coverage uniformity and ligand chemistry emerging as decisive parameters. Although these strategies have enabled notable advances in device longevity under accelerated and outdoor conditions, further efforts are needed to translate these gains into consistent long-term performance.

Outlook and perspective

The engineering of LD perovskites offers wide opportunities for optoelectronic applications, particularly in photovoltaics. By carefully selecting ligand types and compositions, the structure and properties of LD perovskites or perovskitoids can be tuned, thereby imparting tailored optoelectronic functionalities. Such engineering includes designing 3D/LD heterojunction configurations with optimized energetic alignment, orientation, mobility, performance and stability. However, the diversity of techniques and materials currently used impede standardization. Therefore, it is crucial to design and synthesize novel organic ligands that enhance the phase stability and integrity of LD perovskite layers, while ensuring favourable energetic alignment at the 3D/LD heterojunction and charge-extracting interfaces. The primary focus should be on designing monovalent and divalent organic ligands with high pK_a values to stabilize LD perovskites against ion migration and subsequent degradation during device operation.

Ligands containing guanidinium, formamidinium and N-substituted imidazolium moieties are good examples; however, the scope of such ligands can be further extended to include heterocyclic compounds with small ring sizes that can fit into the A-site cationic space. Additionally, attention should be directed towards designing LD organic ligands with a conductive backbone, incorporating either small conjugated systems or aromatic rings to ensure enhanced charge conductivity of the LD layer in both the in-plane and out-of-plane directions. Such ligands may possess divalent ammonium groups that further reduce inorganic sheet distances for improved charge extraction. Nevertheless, these ligands must be sufficiently soluble to ensure the formation of high-quality LD perovskites. The most promising strategy lies in the rational design of organic ligands that can simultaneously stabilize the inorganic framework and enhance charge transport across the 3D/LD junction. For Dion–Jacobson perovskites, divalent ligands with high pK_a values are particularly appealing because they enhance interlayer coupling and suppress ion migration. Expanding beyond conventional diammonium linkers, the exploration of heteroaromatic, bicyclic or rigid aliphatic divalent cations could result in even tighter bonding with Pb–I slabs while still being compatible with solution processing. Concurrently, the incorporation of π-conjugated backbones or short-fused aromatic systems into these ligands can improve carrier delocalization and facilitate vertical charge transport, thereby addressing one of the main limitations of layered perovskites. To ensure high-quality film formation, these ligands must also be engineered with sufficient solubility and steric balance to prevent phase segregation while enabling continuous and homogeneous coverage over large areas. Overall, designing ligands with strong chemical anchoring, electronic conductivity and structural rigidity represents a clear direction for achieving long-term operational stability in phase-pure and homogeneous layered perovskite solar modules.

Advanced characterization techniques are also required to understand the structure and distribution of the LD layer at the surface of the 3D perovskite film. Surface-sensitive and non-destructive measurements are essential for analysing thin LD capping layers. Grazing-incidence X-ray scattering and photoluminescence spectroscopy measurements are commonly used to assess the quality of these layers, providing information on orientation and phase or dimensionality. However, these techniques cannot inform on crystal distribution or whether the capping layer is continuous planar, discontinuous or infiltrated. Therefore, hyperspectral photoluminescence imaging and chemical depth analyses are required. Besides, high-resolution transmission electron microscopy can be used to visualize the LD layer, although there are concerns regarding the susceptibility of the sample to decomposition during measurement and processing. Specific steps are therefore necessary to mitigate beam damage during focused ion beam processing and TEM. Additionally, specific in situ spectroscopy or grazing-incidence X-ray scattering can be used to monitor changes occurring on relevant timescale and length scale. These techniques should also be compatible with large areas to assess the quality of the LD capping layer on module-sized samples.

The variability in stability measurements across different parameters (predominantly under continuous illumination at maximum power-point tracking for hundreds to a few thousand hours) presents a challenge when comparing the efficacy of materials and passivation techniques for achieving optimal performance and stability in PSCs. Increasing the temperature has been proposed to accelerate the degradation and extrapolate data to years of outdoor operation. However, PSCs undergo extensive degradation recovery in darkness and transient dynamics during illumination because of their soft lattice nature, which allows for ion migration and defect repair. Consequently, ageing protocols based on maximum power-point tracking under continuous illumination cannot enable quantitative prediction of outdoor performance. Standardized stability protocols with high predictive power are essential to enable a rapid feedback loop between the development of new materials, interfaces and encapsulation strategies. These protocols should be aligned with industrial stability assessments and conducted expeditiously using accelerated testing methods. Stability tests must also address the unique characteristics of PSCs, incorporating their distinctive recovery dynamics, as these materials may exhibit properties distinct from those of silicon photovoltaics. Recent research works have used the Arrhenius model to analyse stability reports, enabling lifetime predictions under various temperature and light intensity conditions. This approach facilitates a more effective comparison of perovskite cell stability across diverse environmental parameters to predict outdoor stability.

Finally, the pressing need to develop advanced ligands for high-quality, robust LD layers is central for advancing dimensionality engineering strategies for high-performance and stable perovskite solar modules. We anticipate that this Review will serve as a valuable reference for the scientific community, across both academia and industry, in supporting emerging semiconductor optoelectronic applications.

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Author contributions

R.A. conceived the idea, wrote the original draft and contributed substantially to discussion of the content. R.A., D.S.U., Y.L. and S.Z. researched data for the article. R.A. and Y.L. drew the schematics and prepared the figures. All authors wrote the article, reviewed and/or edited the manuscript before submission and during revision.

Competing interests

The authors declare no competing interests.

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